Website Traffic Analysis Report

1. Introduction

The objective of this project was to analyze website traffic data and understand user engagement patterns. The analysis focused on event distribution, traffic trends, geographic distribution, popular content, and conversion behavior. The dataset contained website interaction records including page views, previews, and clicks across multiple countries and cities.

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2. Data Cleaning and Preparation

The dataset was imported into Python using Pandas for analysis. Initial exploration was performed using functions such as head(), info(), and describe().

Data cleaning steps included:

- Checking for missing values.
- Removing duplicate records.
- Converting the date column into datetime format.
- Verifying data consistency across all columns.

The dataset contained minor missing values in country, city, artist, album, and track fields, while a larger number of missing values were observed in the ISRC column. Since the missing values represented a small portion of the dataset, they did not significantly impact the overall analysis.

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3. Exploratory Data Analysis

Event Distribution

The dataset contained three major event types:

- Pageview
- Click
- Preview

Pageviews represented the majority of user interactions, indicating that many users visited content pages. However, only a portion of users proceeded to preview or click actions, showing a drop-off in the engagement funnel.

Traffic Trend Analysis

Daily traffic analysis revealed fluctuations in website activity across different dates. Certain days experienced significantly higher activity levels, indicating periods of increased user engagement.

Geographic Analysis

Traffic originated from multiple countries and cities worldwide.

Key observations:

- The United States generated the highest amount of traffic.
- Several countries contributed moderate traffic volumes.
- Traffic was concentrated in a limited number of major cities.

Artist and Track Performance

Analysis of artist, album, and track data revealed that a small group of artists and tracks generated the majority of user engagement.

Popular content received significantly more pageviews and clicks compared to other entries, suggesting strong audience preferences.

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4. Conversion Analysis

A conversion funnel was created using the event data:

Pageview → Preview → Click

The funnel analysis showed that:

- Many users viewed content pages.
- A smaller percentage moved to preview interactions.
- An even smaller percentage completed click actions.

This indicates opportunities to improve user engagement and conversion rates.

Limitation

The dataset did not contain user identifiers, session identifiers, or session duration information. Therefore, metrics such as sessions, unique users, bounce rate, and average session duration could not be calculated directly.

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5. Key Insights

1. Pageviews accounted for the majority of website interactions.
2. User engagement decreased as visitors moved from pageview to click events.
3. The United States was the leading source of traffic.
4. A small number of artists and tracks generated most of the engagement.
5. Traffic activity varied significantly across dates, indicating peak engagement periods.

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6. Recommendations

1. Improve call-to-action elements to encourage more clicks.
2. Optimize high-traffic landing pages for better user engagement.
3. Focus marketing efforts on top-performing countries and regions.
4. Promote high-performing artists and tracks through featured placements.
5. Conduct A/B testing to improve conversion rates from pageviews to clicks.

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7. Conclusion

This analysis provided valuable insights into website traffic patterns, user engagement behavior, and content performance. The findings indicate that while the website attracts significant traffic, there is an opportunity to improve conversion rates and user engagement through optimization of landing pages, content promotion, and targeted marketing strategies.